

Stewart C. Jamieson

PHD CANDIDATE · AUTONOMOUS ROBOTS & MACHINE LEARNING

☎ (+1) 508 292 2104 | ✉ sjamieson@mit.edu | 🌐 www.stewartjamieson.com | 📷 SJamieson | 🐙 sjamieson | 📄 stewart-jamieson

A roboticist working to develop autonomous systems that co-operate with humans in uncertain, unstructured, and dynamic environments to accomplish their objectives safely and efficiently.

Research Interests & Skills

Artificial Intelligence	Online & Active Learning, Unsupervised Learning, Robust Vision, Spatiotemporal Topic Modelling, AI Ethics
Robotics	Human-Robot Collaboration, Autonomous Exploration, Uncertainty Quantification, Informative Path Planning
Programming	Python, C++20, PyTorch, Keras, ROS, OpenCV, Pandas, Eigen, MATLAB, Mathematica, Bash, Java, Android

Education

Massachusetts Institute of Technology & Woods Hole Oceanographic Institution

Cambridge, MA, USA

PH.D. IN APPLIED OCEAN SCIENCE AND ENGINEERING — MARINE ROBOTICS (5.0/5.0 CGPA)

June 2020 - Present

- Thesis: *Human-Guided Multi-Robot Scientific Exploration in Bandwidth-Limited Environments*
- Developing multi-robot systems to collaboratively explore unfamiliar environments with limited human guidance
- Co-supervised by Dr. Yogesh Girdhar (WHOI) and Prof. Jonathan P. How (MIT)

S.M. IN AERONAUTICS AND ASTRONAUTICS — AUTONOMOUS SYSTEMS (5.0/5.0 CGPA)

June 2018 - May 2020

- Thesis: *Enabling Human-Robot Cooperation in Scientific Exploration of Bandwidth-Limited Environments*
- Co-supervised by Dr. Yogesh Girdhar (WHOI) and Prof. Jonathan P. How (MIT)
- Relevant Coursework: Cognitive Robotics, Visual Navigation for Autonomous Vehicles, Bayesian Modelling & Inference

University of Toronto

Toronto, ON, Canada

B.A.SC. IN ENGINEERING SCIENCE WITH HONOURS — ROBOTICS MAJOR (3.83/4.0 CGPA)

Sept. 2013 - Apr. 2018

- Thesis: *Deep Learning for Robust Vision in Realtime Autonomous Driving*, supervised by Prof. Angela Schoellig
- Applied state-of-the-art uncertainty quantification techniques to achieve safer autonomous driving performance

Work & Research Experience

WHOI's Autonomous Robotics and Perception Laboratory (WARPLab)

Woods Hole, MA, USA

GRADUATE RESEARCH ASSISTANT

June 2018 - Present

- Developing autonomous exploration algorithms for multi-robot teams exploring the deep ocean and coral reefs
- Publications focus on enabling robot co-operation with humans over slow, bandwidth-limited communication channels
- Assisting with the deployment of these novel algorithms into WHOI's world-class deep sea exploration vehicles

aUToronto Self-Driving Car Team (SAE/GM AutoDrive Challenge)

Toronto, ON, Canada

SOFTWARE TEAM LEAD

June 2017 - June 2018

- At the end of my term, aUToronto won **1st place in the SAE/GM AutoDrive Challenge (Year 1)**
- Led a subteam of 12 graduate and undergraduate students working to develop an autonomous Chevrolet Bolt
- My team created the overall system software architecture, sensor drivers, vehicle control interface, and software services

Zebra Technologies Inc.

Mississauga, ON, Canada

SOFTWARE ENGINEERING INTERN, ENGINEERING PRODUCT INNOVATION TEAM

May 2016 - Aug. 2017

- Co-designed and co-developed the software of the first Zebra SmartSight™ prototype, a robot designed for retail operations
- Helped to research and present business applications for robotics, machine learning, and neural networks
- 16 months of C++14 development experience including networking, databases, and high performance computing

Wattpad Inc.

Toronto, ON, Canada

ANDROID SOFTWARE DEVELOPER, READER ACQUISITION TEAM

May 2015 - Sept. 2015

- Wattpad is a worldwide storytelling platform with a community of over 80 million users
- Implemented features designed to attract new users; also implemented A/B tests to validate each features' success

- Searched for, reported, and fixed software bugs in the Android mobile application with over 15 million users
- Designed and implemented a virtual doorman to greet company visitors and notify staff of their arrival

Publications

PEER-REVIEWED CONFERENCE PAPERS

Jamieson, S., How, J. P., & Girdhar, Y. (2023). *DeepSeeColor: Realtime Adaptive Color Correction for Autonomous Underwater Vehicles via Deep Learning Methods*. In 2023 IEEE International Conference on Robotics and Automation (ICRA). London, UK.

Girdhar, Y., McGuire, N., Cai, L., **Jamieson, S.**, McCammon, S., San Soucie, J. E., Todd, J. E., Claus, B., & Mooney, T. A. (2023). *CUREE: A Curious Underwater Robot for Ecosystem Exploration*. In 2023 IEEE International Conference on Robotics and Automation (ICRA). London, UK.

Jamieson, S., Fathian, K., Khosoussi, K., How, J. P., & Girdhar, Y. (2021). *Multi-Robot Distributed Semantic Mapping in Unfamiliar Environments through Online Matching of Learned Representations*. In 2021 IEEE International Conference on Robotics and Automation (ICRA). Xi'an, China.

Jamieson, S., How, J. P., & Girdhar, Y. (2020). *Active Reward Learning for Co-Robotic Vision Based Exploration in Bandwidth Limited Environments*. In 2020 IEEE International Conference on Robotics and Automation (ICRA). Paris, France. **Won Best Paper Award in Service Robotics.**

Girdhar, Y., Cai, L., **Jamieson, S.**, McGuire, N., Flaspohler, G., Suman, S., & Claus, B. (2019). *Enabling Co-Robotic Scientific Exploration of Unknown Environments over a Low Bandwidth Communication Channel*. In 2019 IEEE International Conference on Robotics and Automation (ICRA). Montréal, Canada.

WORKSHOP PAPERS

Yang, D., Cai, L., **Jamieson, S.**, & Girdhar, Y. (2023). *Robot Goes Fishing: Rapid, High-Resolution Biological Hotspot Mapping in Coral Reefs with Vision-Guided Autonomous Underwater Vehicles*. In "CV4Animals" Workshop at CVPR 2023. Vancouver, Canada.

Jamieson, S., Todd, J. E., How, J. P., & Girdhar, Y. (2021). *Communicating Efficiently to Enable Human-Multi-Robot Collaboration in Space Exploration*. In "SpaceCHI: Human-Computer Interaction for Space Exploration" Workshop at CHI 2021. Yokohama, Japan.

Beaulieu, S., Alexander, H., **Jamieson, S.**, Longworth, B., McLean, C., Soenen, K., York, A., Krinos, A., Cai, L., Govostes, R., and Hernandez, C. (2020). *Building a data science curriculum and community for ocean scientists, engineers, and students using The Carpentries model*. In AGU Fall Meeting 2020.

Jamieson, S. (2019). *The Pervasiveness of Deep Learning in Robotics Research Does Not Impede Scientific Insights into Robotics Problems*. In "Debates on the Future of Robotics Research" Workshop at ICRA 2019. Montréal, Canada.

THESES

Jamieson, S. (2020). *Enabling Human-Robot Cooperation in Scientific Exploration of Bandwidth-Limited Environments*. Master's Thesis, Massachusetts Institute of Technology & Woods Hole Oceanographic Institution.

Jamieson, S. (2018). *Deep Learning for Robust Vision in Realtime Autonomous Driving*. B.A.Sc. Thesis, University of Toronto.

Honors & Awards

INTERNATIONAL

2020 **Best Paper Award in Service Robotics (out of 1483 Accepted Papers)**, ICRA 2020

Paris, France

2018 **1st Place Team**, SAE/GM AutoDrive Challenge

Yuma, AZ, USA

ACADEMIC

2018	Dean's Honour List , University of Toronto	<i>Toronto, ON, Canada</i>
2014-16	Dean's Honour List (x3) , University of Toronto	<i>Toronto, ON, Canada</i>
2013	Governor General's Bronze Medal for Academic Excellence , Corpus Christi Catholic Secondary School	<i>Burlington, ON, Canada</i>
2013	Regional Champion , ECOO Programming Competition	<i>Halton, ON, Canada</i>
2010-13	School Champion (x4) , Waterloo CEMC Math Contest	<i>Burlington, ON, Canada</i>

Presentations

Videos and other materials used in some of the following presentations are available at www.stewartjamieson.com

ROBOTS: How new technologies are advancing ocean research, exploration, and science-based solutions	<i>Online</i>
PRESENTER, WHOI OCEAN ENCOUNTERS (SEASON 5 EPISODE 4)	<i>May 2023</i>
Multi-Robot Reef Monitoring, Mapping, & Human-Guided Exploration	<i>Online</i>
INVITED TALK, NCCOS SfM/AI WORKING GROUP	<i>March 2023</i>
Rapid, Comprehensive Coral Reef Monitoring with AUVs	<i>Bremen, Germany</i>
INVITED TALK, 15TH INTERNATIONAL CORAL REEF SYMPOSIUM	<i>July 2022</i>
Communicating Efficiently to Enable Human-Multi-Robot Collaboration	<i>Online</i>
POSTER PRESENTATION, SPACECHI WORKSHOP @ CHI 2021	<i>May 2021</i>
Human-Robot Cooperation for Exploring Bandwidth-Limited Environments	<i>Woods Hole, MA, USA</i>
INVITED TALK, WHOI AOPD DEPARTMENT SEMINAR SERIES	<i>July 2020</i>
Active Reward Learning for Co-Robotic Exploration in Bandwidth-Limited Environments	<i>Cambridge, MA, USA</i>
INVITED PRESENTER & PANELIST, ICRAxMIT	<i>June 2020</i>
Deep Learning Does Not Impede Scientific Insights into Robotics Problems	<i>Montréal, QC, Canada</i>
INVITED LIGHTNING TALK, DEBATES ON THE FUTURE OF ROBOTICS RESEARCH, ICRA 2019	<i>May 2019</i>
16.412 Lecture: Multi-Robot Adaptive Sampling	<i>Cambridge, MA, USA</i>
CO-LECTURER, MASSACHUSETTS INSTITUTE OF TECHNOLOGY	<i>Apr. 2019</i>
An Introduction to Neural Networks and Machine Learning	<i>Mississauga, ON, Canada</i>
LEAD PRESENTER FOR ZEBRA TECHNOLOGIES INC. "LUNCH & LEARN" (200+ ATTENDEES)	<i>Jan. 2017</i>
Should Robots Have Rights?	<i>Toronto, ON, Canada</i>
CO-PRESENTER IN DEBATE AT THE UNIVERSITY OF TORONTO	<i>Dec. 2015</i>
A Customized Graphical Checklist for Efficient Ambulance Inventory	<i>Toronto, ON, Canada</i>
CO-PRESENTER IN "PRAXIS II SHOWCASE" AT THE UNIVERSITY OF TORONTO	<i>Apr. 2014</i>

Teaching

Massachusetts Institute of Technology	<i>Worldwide</i>
TEACHING ASSISTANT	<i>Aug. 2020 - Dec. 2020</i>
• Fall 2020: 16.485 Visual Navigation for Autonomous Vehicles (Prof. Luca Carlone)	
The Carpentries	<i>Worldwide</i>
CERTIFIED SOFTWARE CARPENTRIES INSTRUCTOR	<i>Oct. 2019 - Present</i>
• June 2021: Co-Instructor for WHOI Student Python Workshop Series 2021	
• Oct. 2020: Helper for WHOI Data Carpentry Workshop Fall 2020	
• July 2020: Co-Instructor for WHOI Student Python Workshop Series 2020	
• Oct. 2019: Helper for WHOI Software Carpentry Workshop Fall 2019	

Academic Service

- Organizer of:
 - AI for Earth & Space Science Workshop @ ICLR 2022
- Reviewed Journal Submissions for:

- Robotics and Automation Letters (RA-L)
- Journal of Aerospace Information Systems (JAIS)
- Reviewed Conference Submissions for:
 - International Conference on Learning Representations (ICLR)
 - Conference on Neural Information Processing Systems (NeurIPS)
 - International Conference on Machine Learning (ICML)
 - International Conference on Robotics and Automation (ICRA)
 - International Conference on Intelligent Robots and Systems (IROS)

Professional Service

MIT-WHOI Joint Program

Woods Hole, MA, USA

ELECTED AT-LARGE REPRESENTATIVE

Oct. 2020 - Oct. 2021

- Co-hosted monthly online student events to maintain social cohesion during worldwide pandemic
- Developed and published the MIT-WHOI Joint Program online photoboard

Zebra Technologies Inc.

Mississauga, ON, Canada

EDITOR, EMC INNOVATION NEWSLETTER

May 2016 - Aug. 2017

- Edited bi-monthly department newsletter and distributed it to over 1700 engineers
- Commissioned, reviewed, and published articles about recent trends and innovations in electronics, robotics, etc.

Graduate Student Member

Worldwide

IEEE (INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS)

Nov. 2013 - Present

Personal Interests

Corpus Christi Jazz Horns and Concert Band

Burlington, ON, Canada

SAXOPHONIST

Sept. 2009 - June 2013

- Performed in the Atlantic Music Festival (2013), Toronto Music Festival (2012)